

HealthConnect Project Summary

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Dataset: HealthConnect Appointment Dataset

Review of Week 5 Plan

Complete Data Preparation

- Replace all missing values and validate fields.
- Document all data transformations.

Detailed Exploratory Data Analysis

- Create summary statistics for all variables.
- Build visualizations exploring each business question.

Identify patterns and trends in no-show behavior.

- Statistical Analysis
- Outline hypotheses.

Identify which variables will be tested.

- Dashboard development
- Define Dax measures needed.
- Design the executive summary.

Achieved Goals

- All data preparation tasks were completed, including handling missing values, validating fields, and creating new features.
- A comprehensive EDA was performed, exploring all the proposed business questions through visualizations and analysis.
- Performed statistical analysis for numerical fields and outlined hypothesis
- All 6 KPIs were defined, calculated, and interpreted.
- Developed interactive executive dashboard and KPI's alongside their DAX measures

Key Findings

- Ladies show increase in no-show rate between and decrease when they reach 65+ while man on the other hand shows slight decline in no-show rate as they grow older.
- The no-show rate is high on Sundays and Fridays.
- The percentage of no-shows increases drastically with the number of previous no-shows
- No-show rate increases with distance up to 40km when a reminder is sent, with a sharp spike at 30-40km (77%), then drops at 40km+.

- Email is the most used channel but has a no-show rate of 48%, making it the second least effective channel after WhatsApp.

Major Challenges Encountered

Challenges

- Differentiating between true 'null' values and the 'none' value in the reminder_channel column.
- Sorting days of the week in Power BI to be in chronological order

Solutions

- I resolved this by closely re-examining the data dictionary, which clarified that 'none' was an intentional value meaning 'No Reminder' was sent. This insight prevented me from inputting a value incorrectly and allowed me to create a meaningful category.
- Created date table with 'day' and 'day order' column, which I used to create a relationship between appointment_day and day from date table ordering my days in chronological order

Important Decisions Made

- Created 'distance_group' column making it easier to visualize distance trend
- Created 'No reminder' category for reminder channel instead of treating it as null value, this decision was made to accurately represent the data and allow for a proper comparison of the effectiveness of different reminder channels.
- Created 'Month' column for easier analysis of periodical trends
- Modelled my visuals in python before creating Power BI dashboards this enabled me to differentiate visuals for executive dashboards and other report pages

Changes to Proposed Week 5 Approach

The outlined plan was followed without any alterations

Cross Track Collaboration Completed

Collaborated with Project Management and Data Science tracks. PM helped identify No-Show Rate and Show Rate as primary success metrics. Data Science on the other hand decided to add distance × reminder interaction to their model based on my findings and on I decided to keep my waiting time analysis though its only used for historical analysis

Remaining Work

- Create other dashboards
- Begin deeper collaboration to prepare for Week 6, focusing on integrating the data analytics findings with the data science and ML engineering solutions.
- Document findings and making improvements

Proposed Week 6 Approach

- Dig deeper into the analysis by creating dedicated dashboards
- Test hypothesis
- Refine dashboards to accommodate change
- Present insights to other tracks and note any improvements